

A comparison of preferential flow characteristics and influencing factors between two soils developed in the karst region of Southwest China

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ABSTRACTS

In this study, differences in the characteristics of the preferential flow of water in characteristics of two typical rocky soils in southwestern karst areas were investigated and the factors driving these differences were identified. Bright blue dye tracers and image processing techniques were used to determine the morphological characteristics of preferential flow in dolomite and sandy shale soils, and the relationships between environmental factors and preferential flow characteristics were quantitatively evaluated. The results showed that (1) the depth of homogenous matrix flow in karst dolomite soil was significantly shallower than that in sandy shale soil, and preferential flow was highly differentiated, mostly into branches and patches, and was dominated by vertical extension. Preferential flow in sandy shale soil clearly occurred via lateral infiltration, mostly occurring in clusters and blocks, with a low degree of differentiation. (2) The dolomite and sandy shale soils were dominated by dyed paths with widths of $W < 20$ mm, with the number of dyed paths of each width in the dolomite soils being greater than those in the sandy shale soil, and the water flow type in both soils was dominated by highly interactive macroporous flow, with percentages in the two soils ranging from 55.11 % to 68.48 % and from 13.37 % to 67.83 %, respectively. (3) Both the plant root distribution and soil properties affected the dyed area ratio and the distribution of dyed paths, with dyed paths with a widths of $W > 200$ mm being the most significantly affected. Redundancy analyses revealed that the main factors influencing preferential flow were the organic matter content in the dolomite soils, with a contribution of 69.10 %, and the root surface area density in the sandy shale soil, with a contribution of 77.20 %. In summary, changes in the soil properties and plant roots distribution caused by lithological differences in karst areas were important factors affecting the characteristics of preferential flow.

1. Introduction

Water infiltration through soil is an important process that links groundwater, surface water, and soil water (Qiu et al., 2023; Wang et al., 2008; Zhang et al., 2019). Preferential flow, a common form of soil moisture movement, causes water and solutes to bypass parts of the soil matrix, preferentially pass through large pore spaces with high hydraulic conductivity, and rapidly move to deeper layers of soil to recharge groundwater (Clothier et al., 2007; Gerke, 2010; Jarvis, 2020). However, preferential flow can cause the rapid loss of soil nutrients as well as water (Isensee et al., 1988), and it is now a key issue in the studies of soil erosion, groundwater contamination, and geohazards

(Allaire et al., 2009; Wu et al., 2016; Bachmair et al., 2009; Huang et al., 2017). Studies on preferential flow and related topics have led to the rapid introduction of soil moisture studies into the field of non-homogeneity research (Hlaváčiková et al., 2019; Jarvis et al., 2007). However, many environmental factors affect preferential flow characteristics, and regional differences exist; therefore, conducting a quantitative study of the relationships between the characteristics of preferential flow in soil and related environmental factors is a major challenge.

The karst landscape in Southwest China is one of three major karst areas in the world, and it accounts for 15 % of the global karst area and nearly one-third of China's land area (Guo et al., 2013). This area has the

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large rock outcrops, strong karst development, and acute human-land conflicts (Herman et al., 2012), and the ecological environment is very fragile (Jiang et al., 2014). The lack of water resources and the uneven spatial and temporal distributions of water limit the sustainable development of karst areas, and systematic studies of soil hydrological processes and related factors are necessary for water resource protection. However, the lithologies of karst areas are complex, and carbonate and noncarbonate parent rocks are often distributed in alternating bands. There are many differences in soil properties and structures among soils of different lithologies. Due to the strong karst activity in carbonate rock areas, especially those containing dolomite, which is a widely distributed pure carbonate rock, soils with low acid-insoluble contents form (Wang et al., 2010). Such soils often feature an upper layer of highly permeable soil overlying bedrock, and the rock-soil interface is distinct. Such areas are the key zones for subsequent rocky desertification management (Jiang et al., 2014). In noncarbonate areas, sandy shale is most typical, the soil layer is relatively deep and continuous, pipelines along fractures are mostly undeveloped, and the vertical soil profile is not significantly differentiated. However, whether these soil differences directly lead to changes in the water transport characteristics and the factors driving water flow are still unknown.

In previous studies, scholars have investigated karst areas and confirmed the widespread occurrence of preferential flow in these areas (Liu and She, 2020; Wang et al., 2018). However, most of these studies have focused on the impact of exposed bedrock on soil permeability in carbonate rock-derived soils (Zhao et al., 2020), the characteristics and trends of water transport at the rock-soil interface (Sohrt et al., 2014), and the influence of the morphology and spatial heterogeneity of cracks in soil on water infiltration rates (Liu and She, 2020; Wang et al., 2018). However, few studies have assessed the impact of lithological differences within a given area on the characteristics of water transport in soil, quantified the degree of preferential flow occurrence as well as comprehensively analyzed the relationship between preferential flow distribution characteristics and external influences. In addition, the population in the karst region of Southwest China has reached 100 million, and land reclamation for human activities has had a severe impact on the already fragile environment of the region (National Development and Reform Commission, 2020). Moreover, due to the limitations of environmental factors, the problems of rocky desertification and poverty have been intertwined for a long time, and the contradiction between its ecological protection and social development is prominent (Jiang et al., 2014). Therefore, there is an urgent need to understand the characteristics of soil water infiltration in two types of rocky soils and their relationships with environmental factors for the ecological restoration and reconstruction of karst areas.

It is difficult to directly measure preferential flow because of the heterogeneity and instability of soil during moisture transport. The dye tracer method is a common method for assessing the extent to which preferential flow occurs and for tracking flow patterns (Flury and Flühler, 1994; Ghodrati and Jury, 1990; Kodešová et al., 2012; Mooney and Morris, 2008). Therefore, this study investigated various typical land use types, such as cultivated land, orchard, shrub grassland and secondary forest, on land dominated by dolomite and sandy shale in the southwestern karst region of China. We used dye tracer experiments and image analysis techniques combined with analyses of the distributions of plant root systems and soil physical and chemical properties to (i) quantify and analyze the occurrence of preferential flow in the two types of typical rocky soils and the morphological characteristics of flow in the karst region and (ii) elucidate similarities and differences in the main factors influencing water infiltration in the two typical rocky soils in the study area to provide scientific support for the comprehensive management of rocky desertification in the study area, as well as for the systematic utilization of regional water resources.

2. Materials and methods

2.1. Overview of the research area

The research area is located in the Mulian Comprehensive Experimental Demonstration Zone of Ecosystem Observation of the Chinese Academy of Sciences (24°43'58.9" N, 108°18'56.9" E), Huanjiang County, Hechi city, Guangxi, and it contains a typical karst landscape unit, as shown in Fig. 1. The altitudes are between 272.0 and 647.2 m, and nonzonal soil and zonal soil are present in the area. It has a subtropical monsoon climate and an average annual temperature of 19.9°C. The region is warm and wet, with an average annual rainfall of up to 1389 mm; however, the spatial and temporal distributions of precipitation are uneven, and rainfall is mainly concentrated from April to September, accounting for 74 % of the annual rainfall. October to March is the dry season, and seasonal drought and flood disasters occur frequently.

2.2. Sample site selection and sample collection

In this experiment, four typical land use types, namely, cultivated land (CL), orchard (OR), scrub grassland (SG) and secondary forest (SF), developed on dolomite and sandy shale in the karst region were taken as the research objects. To ensure that the site conditions and geographical environments of each sample plot were basically consistent, a full investigation was conducted. The basic distribution of the sample plots is shown in Table 1. In this experiment, 20×20 m quadrats were divided into various plots, the tracer sites were selected following an "S" shape, and points near the sites (approximately 30 cm) were selected. Scattered soil samples and cutting ring soil samples were collected from the top 10 cm layer, and 3–5 sampling points were selected for each layer of soil with a collection depth of 50 cm. The collected loose soil was mixed to determine basic physical and chemical properties. The soil organic matter content was determined via an external heating method with potassium dichromate (Pan et al., 2017). The pipette method was used to determine the soil particle composition, and the soil particles were categorized as sand (2.0–0.05 mm), silt (0.05–0.002 mm), or clay (<0.002 mm), according to the United States Department of Agriculture (USDA) Soil Taxonomy. The soil bulk density and total porosity were measured by the cutting ring method, and the soil moisture was determined by a drying method (Lu, 2000). The basic physical and chemical properties of the tested soil are shown in Table 2.

2.3. Preferential flow dyeing and tracing

In the sample plot, 3–5 dyeing tracer test sites were set up at locations with relatively gentle terrain and no obvious water accumulation. To reduce the influence of major plant roots, a distance of more than 100 cm was maintained, the soil surface was levelled, and the dyeing area was divided. A 120×70×30 cm iron frame was driven into the soil approximately 15 cm deep, and the soil attached to the iron frame was compacted to avoid infiltration of the bright blue dye along the cracks. The number of observed profiles at various points was at least 5.

In this study, a local cumulative rainfall of 40 mm over 24 hours was used as the standard to observe the characteristics of soil preferential flow, and the bright blue dyeing concentration was set to 4 g/L. A uniform spraying process was used to ensure that there was no significant water accumulation or runoff on the surface. The soil was covered with a plastic film immediately after the dye was sprayed. After 24 hours, the iron sheet frame and the rainproof material were slowly removed, the vertical and horizontal sections of the soil were excavated, and images of each section were taken with the help of a Canon SLR camera (1200 W), as shown in Fig. 2.

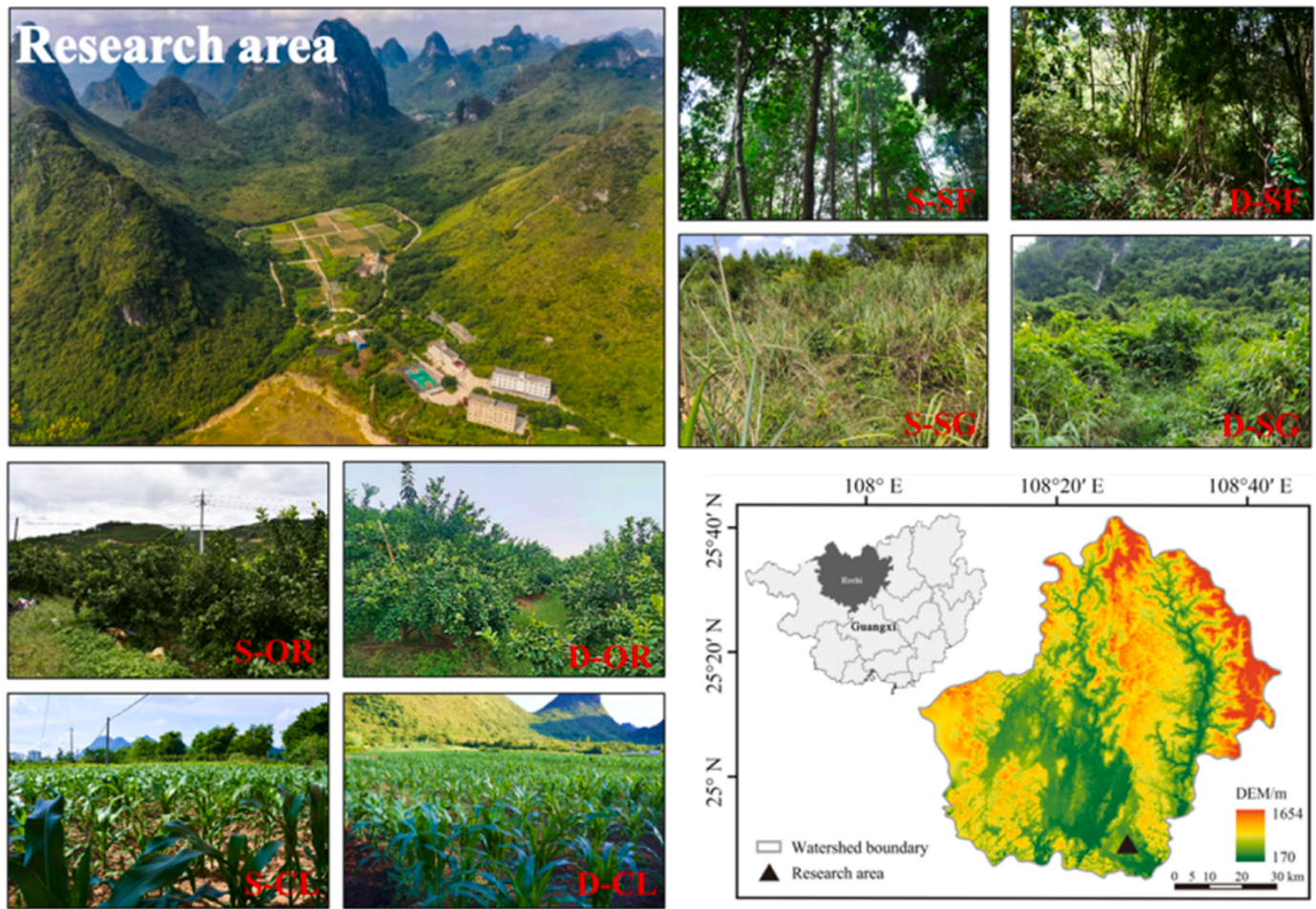


Fig. 1. Location of the study area. Note: CL refers to arable land, OR to orchards, SG to scrub grassland, SF to secondary woodland, below; S- refers to the sandy shale development soil sample site and D- refers to the dolomite development soil sample site. Same as below.

2.4. Analysis of dyed profile images

Due to the lens distortion of the optical lens of the camera, there is a gap between the dyed profile image and the actual image; therefore, it was necessary to use PS software for image processing and then use MATLAB 2017 software to quantify the image and obtain the image feature parameters. Fig. 3 shows the operation flowchart.

- (1) The soil profile dyed area ratio (SDA) (%) is the ratio of the dyed area to the total area in the entire soil profile (Flury et al., 1994) and was calculated as follows:

$$DC = \left(\frac{D}{D + ND} \right) \times 100\% \quad (1)$$

In the above equation, D represents the total area of preferential flow dyeing in the vertical profile (cm^2), ND represents the sum of the total area of the undyed part in the profile (cm^2), and the depth and width of the soil profile are 50 cm.

(2) Matrix flow depth (UniFr, cm): When the dyed area ratio in the vertical soil profile decreases to 80 %, the depth of the soil layer is the matrix flow depth, and the part above this depth is the matrix flow area (Mei et al., 2018).

(3) The dyeing path width (DPW, mm) is the actual width of each independent dyed path in the corresponding soil layer, which characterizes the flow pattern of water in the soil and the degree of preferential flow differentiation (Janssen and Lennartz, 2008). The formula for calculation is as follows:

$$DPW = \frac{S}{R} \times 100\% \quad (2)$$

In the formula, S represents the number of independent dyed path pixels (pixels), R represents the resolution of the pixels ($1 \text{ pixel} \cdot \text{mm}^{-1}$) and the types of water flowing through the soil are divided based on DPW diameter class and occupied area (Weiler and Flühler, 2004) (Table 4).

2.5. Determination of the vegetation root distribution

After the excavation and imaging of the horizontal profile, four massive $10 \times 10 \text{ cm}$ soil samples were randomly selected from each profile and soaked to separate the roots from the soil. The fresh roots were collected and washed under natural light, and the root samples from various sites were scanned using an Epson EU-88 (EPSON, Japan) scanner. WinRHIZO software was used to quantify the collected roots and calculate parameters such as the root length density, root surface area and root volume (Fig. 4). The results are detailed in Table 3.

2.6. Data analysis

The experimental data obtained in this study were preliminarily sorted in Excel 2019. MATLAB 2017 was used to extract, calculate and plot the feature values of the dyed profile. SPSS 21.0 was used for statistically analyze the data, and the tests included the normality test, homogeneity test of variance, and one-way analysis of variance (ANOVA) of the soil properties and root parameters among the different

Table 1
Basic situation of the study plot.

Lithology	Sample land type	Latitude	Longitude	Altitude/(m)	Main vegetation types
Dolomite	CL	24°76'38"	108°31'68"	208.8	<i>Zea mays</i> ; <i>Euphorbia hirta</i> ; <i>Xanthium strumarium</i>
	OR	24°74'33"	108°32'99"	302.6	<i>Citrus reticulata</i> ; <i>Gynostemma pentaphyllum</i> ; <i>Erigeron canadensis</i> ; <i>Sphagneticola calendulacea</i> ; <i>Bidens pilosa</i> var. <i>radiata</i>
	SG	24°73'68"	108°32'80"	286.8	<i>Vitex negundo</i> ; <i>Cipadessa cinerascens</i> ; <i>Cayratia albifolia</i> ; <i>Sabia swinhoei</i>
	SF	24°73'68"	108°32'65"	301.8	<i>Alangium platanifolium</i> ; <i>Acer coriaceifolia</i> ; <i>Moringa oleifera</i> ; <i>Cyclosorus dentatus</i> ; <i>Jasminum pentaneurum</i>
Sandy shale	CL	24°73'58"	108°32'94"	293.0	<i>Zea mays</i> ; <i>Galinsoga quadradiata</i> ; <i>Digitaria sanguinalis</i>
	OR	24°74'50"	108°32'97"	274.5	<i>Citrus reticulata</i> ; <i>Setaria viridis</i> ; <i>Lactuca serriola</i> ; <i>Spermacoce alata</i>
	SG	24°74'09"	108°33'22"	289.6	<i>Rhus chinensis</i> ; <i>Miscanthus floridulus</i> ; <i>pericampylus glaucus</i> ; <i>Dicranopteris pedata</i> ; <i>Embelia ribes</i>
	SF	24°74'16"	108°33'19"	289.7	<i>Ficus hirta</i> ; <i>Duranta erecta</i> ; <i>Smilax davidiana</i> ; <i>Melicope pteleifolia</i> ; <i>Cinnamomum camphora</i> ; <i>Blechnum orientale</i>

Note: CL refers to cultivated land, OR to orchards, SG to scrub grassland, SF to secondary forest land. The same below.

land uses and lithologies. Additionally, the least significant difference (LSD) test method was used for multiple comparisons of significance (significance level was set at $p < 0.05$). Canono 5 software was used for redundancy analysis, and Origin 2023 software was used to perform linear fitting, correlation analysis, and mapping of the data.

3. Results and analysis

3.1. Morphological characteristics of preferential flow in soil

The binary preferential flow characteristics of the vertical profiles of the two rocky soils in the karst area are shown in Figs. 5 and 6. The white area is the undyed area, and the black area is the dyed area. In the surface soil, the dyed area is relatively uniform, and the area accounting

for more than 80 % of the total area is called the uniform flow area. Areas with smaller dyed areas are the preferential flow areas, and undyed areas are matrix flow areas (Hagedorn and Bundt, 2002). The results showed that there were significant differences between the preferential flow characteristics of the soils in the dolomite and sandy shale sampling sites. Dolomite soil preferential flow dyeing paths in the samples formed with increasing depth in the soil layer, which was characterized by sharp differentiation. The preferential flow morphology was mainly in the form of dendrites and a small number of patches, with vertical extension of the main layer. In particular, differentiation was most common in the secondary forestland (SF) and shrub grassland (SG), followed by that in the orchard (OR), and differentiation was least common in the cultivated land (CL). The preferential flow of soil water during infiltration was more obvious in the dolomite soil than in the sandy shale soil, with a greater degree of morphological differentiation.

The soil preferential flow in soil paths in sandy shale tended to have greater widths and significantly lower numbers than those in dolomite, with a lower degree of morphological differentiation (Fig. 6). Among them, the dyed areas were mostly clumpy or blocky, the number of dyed paths was low, and the width was large. Except in the SF, the paths were concentrated in the 10–35 cm soil layer, and obvious lateral infiltration occurred during the infiltration process. The preferential paths in the vertical profile in the plots of the SF samples mostly showed patchy and partly elongated morphologies, with good and uniform connectivity. The paths were directly connected to the bottom of the layer profile, and the differentiation of preferential flow morphology was more obvious. This indicated that although the soil parent material for the sites was the same and that differences in land use and influences from soil properties led to differences in the preferential flow pattern of the soil in the profile. Overall, the water infiltration capacity of the sandy shale soils was significantly lower than that of the dolomite soils, which may be related to differences in the spatial structure of the two rocky soils.

3.2. Distribution characteristics of soil preferential flow in soil

The dyed area ratio (DAR) reflects the infiltration of soil water at different depths, and the larger the value is, the more water infiltration occurs, revealing the rapid relocation of the wetting front (Aeby et al., 1997). As shown in Fig. 7 and Table 5, the DARs of both soils were greater than 80 % in the surface layer, and with increasing soil depth, the DARs of the two soils exhibited differences, which indicated nonuniform infiltration and preferential flow differentiation. Overall, under an equivalent water supply, the average DAR and total area of the dyed profile were greater in the dolomite profiles than in the sandy shale soils, and the depth to substrate flow was significantly less in the OR, SG, and SF ($p < 0.05$). In addition, the preferential flow DAR curves of the dolomite soils exhibited an "S" shape, especially in OR, SG and SF, and were distributed throughout the entire soil profile, while the DAR of the sandy shale soil was relatively smooth, except in SF; the DAR curves of the other samples were distributed in the 0–35 cm profile layer. This finding indicates that the range of soil water movement in dolomite soil was greater than that in sandy shale soil, the vertical migration and infiltration capacity of soil water were greater, and the differentiation of the dyeing path occurred more rapidly in dolomite soil than in sandy shale soil.

Change in land use also affected the characteristics of preferential flow. The DAR of dolomite CL decreased with soil depth, dyed paths appeared in the 10–20 cm soil layer, and the OR, SG, and SF DARs showed clear "S"-shaped trends. These results indicated that lateral infiltration of water induced morphological differences in the dyed pathways. Among the dolomite soils with different land use types, the trend in the total dyed area of the profile was $SF > OR > SG > CL$, with significant ($p < 0.05$) differences among all samples. Sandy shale CL produced lateral transport at depths of 5–7 cm, dyed patterns began to differentiate (Fig. 7), and herbaceous root distributions were generally

Table 2
Basic physical and chemical properties of soil.

Lithology	Sample land type	Depth of soil layer/(cm)	Soil moisture content	Soil bulk density/(g·cm ⁻³)	Sand/(%)	Silt/(%)	Clay/(%)	Total porosity / (%)	Organic matter/(g·kg ⁻¹)
Dolomite	CL	0–10	29.60±0.03a	1.20±0.05c	25.50 ±0.74ab	35.66 ±1.93a	38.84 ±1.19b	60.98±4.46a	48.13 ±3.31a
		10–20	27.25±0.02a	1.25±0.04bc	25.08 ±1.69ab	33.16 ±0.84ab	41.76 ±0.85ab	57.87±1.01ab	19.07 ±0.36b
		20–30	25.79±3.96ab	1.27±0.05abc	26.15 ±0.67a	29.55 ±0.34b	44.30 ±1.01ab	57.84±3.32ab	11.14 ±1.67c
		30–40	19.69±0.20b	1.30±0.04ab	26.64 ±0.14a	29.02 ±0.03b	44.34 ±0.17ab	56.08±2.05ab	8.59±0.93c
		40–50	19.66±0.10b	1.35±0.02a	21.31 ±1.60b	31.32 ±1.76ab	47.37 ±3.36a	51.27±2.84b	6.07±0.80c
	OR	0–10	27.16±1.50ab	0.95±0.02c	22.11 ±2.48a	24.89 ±1.63b	53.00 ±2.65ab	60.30±1.04a	31.84 ±0.69a
		10–20	28.36±0.35a	1.11±0.05ab	19.41 ±3.29ab	24.50 ±2.50b	56.09 ±0.78ab	60.48±2.64a	28.50 ±1.40b
		20–30	25.46±0.84ab	1.09±0.05b	18.02 ±1.23ab	26.16 ±0.53ab	55.82 ±1.76ab	56.14±2.40ab	26.67 ±0.22b
		30–40	23.98±0.28bc	1.16±0.03ab	21.07 ±4.28a	31.02 ±1.82a	47.91 ±2.10b	54.73±2.38b	25.86 ±0.20b
		40–50	22.61±1.41c	1.17±0.01a	16.68 ±2.40b	25.17 ±0.96ab	58.15 ±3.36a	52.62±1.73b	11.18 ±0.81c
	SG	0–10	30.04±0.08ab	0.87±0.03b	20.68 ±2.97b	33.93 ±4.01a	45.39 ±1.05c	63.65±0.53a	55.87 ±0.96a
		10–20	30.45±0.23a	1.08±0.02ab	28.26 ±1.25a	20.49 ±0.40b	51.25 ±0.85b	63.17±1.26a	38.78 ±2.24b
		20–30	28.74±0.79ab	1.07±0.03ab	20.40 ±2.12b	22.08 ±1.83b	57.52 ±2.29a	62.78±1.01b	31.54 ±1.44c
		30–40	28.79±0.51ab	1.09±0.02ab	18.81 ±1.43b	24.91 ±3.10b	56.28 ±2.33ab	61.70±2.57b	18.49 ±0.01d
		40–50	26.94±1.55b	1.22±0.17a	19.33 ±0.02b	23.03 ±0.32b	57.64 ±0.34a	61.40±1.15b	4.44±0.77e
	SF	0–10	30.39±0.48a	0.98±0.07b	34.66 ±2.93a	22.56 ±2.42ab	42.78 ±2.65c	63.53±1.83a	78.96 ±0.36a
		10–20	30.54±0.16a	1.09±0.03a	23.95 ±2.54b	23.80 ±1.58a	52.25 ±0.96b	68.46±5.13a	54.66 ±2.95b
		20–30	29.86±0.48a	1.13±0.03a	21.25 ±2.00b	21.66 ±4.64ab	57.09 ±2.64ab	66.55±2.11ab	52.89 ±2.34b
		30–40	28.29±0.62a	1.10±0.04a	23.34 ±1.35b	15.25 ±1.36c	61.41 ±2.24a	63.55±3.01b	14.07 ±4.54c
		40–50	25.32±1.04b	1.11±0.02a	18.69 ±1.11b	17.87 ±0.02bc	63.44 ±1.13a	56.99±2.97c	9.88±0.33c
Sandy shale	CL	0–10	23.27±0.09b	1.17±0.08b	46.89 ±0.22a	24.63 ±0.24b	28.48 ±0.46ab	57.68±1.89a	24.59 ±1.45a
		10–20	28.90±0.17a	1.24±0.02b	39.73 ±0.17b	26.24 ±2.42b	34.03 ±2.25a	47.62±0.65b	21.81 ±1.39ab
		20–30	23.37±0.46b	1.50±0.04a	44.33 ±1.46ab	32.80 ±4.45a	22.87 ±2.99b	46.58±1.89b	15.66 ±1.37bc
		30–40	21.98±0.57bc	1.50±0.06a	43.16 ±2.02ab	29.55 ±1.66ab	27.29 ±0.36ab	44.66±0.90b	10.17 ±4.07c
		40–50	19.54±1.88c	1.55±0.02a	42.16 ±1.96ab	27.39 ±0.82b	30.45 ±2.78ab	45.27±2.38b	8.03±1.17c
	OR	0–10	15.89±0.50a	1.23±0.02b	64.76 ±0.43a	14.53 ±0.38b	20.71 ±0.81c	58.89±1.17a	23.65 ±2.29a
		10–20	14.61±2.24ab	1.52±0.05a	60.73 ±2.29ab	14.48 ±0.93b	24.79 ±1.36b	50.92±3.12b	16.92 ±1.49ab
		20–30	13.11±0.67ab	1.53±0.05a	61.54 ±4.33a	14.41 ±0.33b	24.05 ±0.00b	47.97±2.61bc	13.64 ±2.59b
		30–40	9.97±1.26bc	1.52±0.02a	65.57 ±0.29a	10.66 ±0.43c	23.77 ±0.14b	43.65±1.88c	10.87 ±0.82b
		40–50	7.31±0.85c	1.57±0.05a	53.12 ±0.43b	18.37 ±1.02a	28.51 ±0.59a	32.01±4.83d	10.68 ±3.20b
	SG	0–10	23.67±1.38a	1.14±0.02b	59.31 ±0.80a	19.39 ±0.51b	21.30 ±1.30b	54.89±1.09a	26.17 ±0.22a
		10–20	22.20±0.20a	1.27±0.06ab	55.97 ±1.33ab	15.42 ±0.61b	28.61 ±1.94a	53.70±2.64a	16.96 ±1.53b
		20–30	19.22±0.33b	1.29±0.06ab	58.21 ±2.86a	16.88 ±1.56b	24.91 ±1.30ab	50.28±5.40b	14.89 ±3.84bc
		30–40	18.59±0.22b	1.33±0.11a	54.36 ±1.31ab	19.74 ±1.61b	25.90 ±2.92ab	50.33±2.82b	14.49 ±2.80bc
		40–50	19.46±0.17b	1.33±0.09a	47.69 ±4.12b	28.11 ±1.02a	24.20 ±0.14ab	48.12±2.60c	6.71±0.76c
	SF	0–10	17.93±0.05a	1.18±0.03b	48.88 ±1.70b	22.69 ±1.90a	28.43 ±0.20c	62.92±2.65a	33.03 ±2.71a

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Table 2 (continued)

Lithology	Sample land type	Depth of soil layer/(cm)	Soil moisture content	Soil bulk density/(g·cm ⁻³)	Sand/(%)	Silt/(%)	Clay/(%)	Total porosity/(%)	Organic matter/(g·kg ⁻¹)
		10–20	17.09±0.30a	1.21±0.02b	56.00±0.86a	13.75±0.14b	30.25±0.71bc	60.51±1.95a	24.92±0.19b
		20–30	16.64±0.65ab	1.22±0.03b	51.88±2.08zb	18.24±0.91ab	29.88±1.17bc	59.91±3.88b	23.44±4.49bc
		30–40	15.97±0.91ab	1.27±0.03ab	52.63±1.22ab	14.03±2.23b	33.34±1.00ab	53.89±1.03c	17.22±0.11 cd
		40–50	14.77±1.48b	1.34±0.11a	52.12±0.10ab	12.66±.32b	35.22±1.43a	50.05±1.16c	14.22±1.22d

Note: Lowercase letters indicate significant differences in average total dyed area among land use types ($p<0.05$). The numerical arrangement in the table is mean ± standard deviation. The same below.

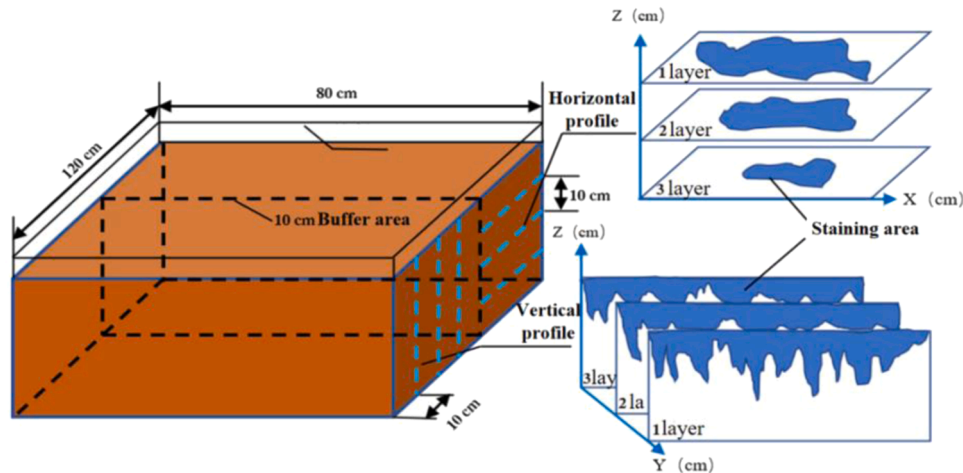


Fig. 2. Excavation diagram of dyed tracer soil profile.

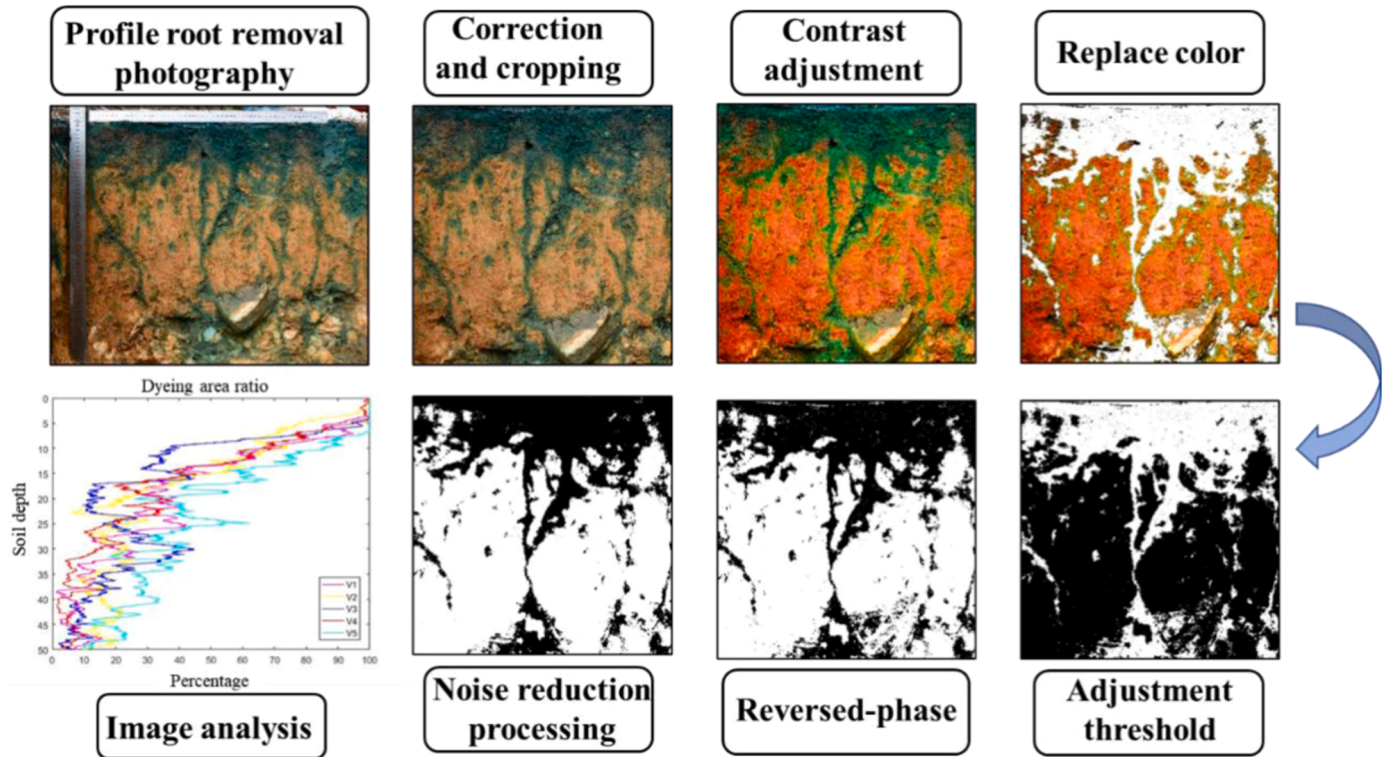


Fig. 3. Flow chart of soil profile image processing.

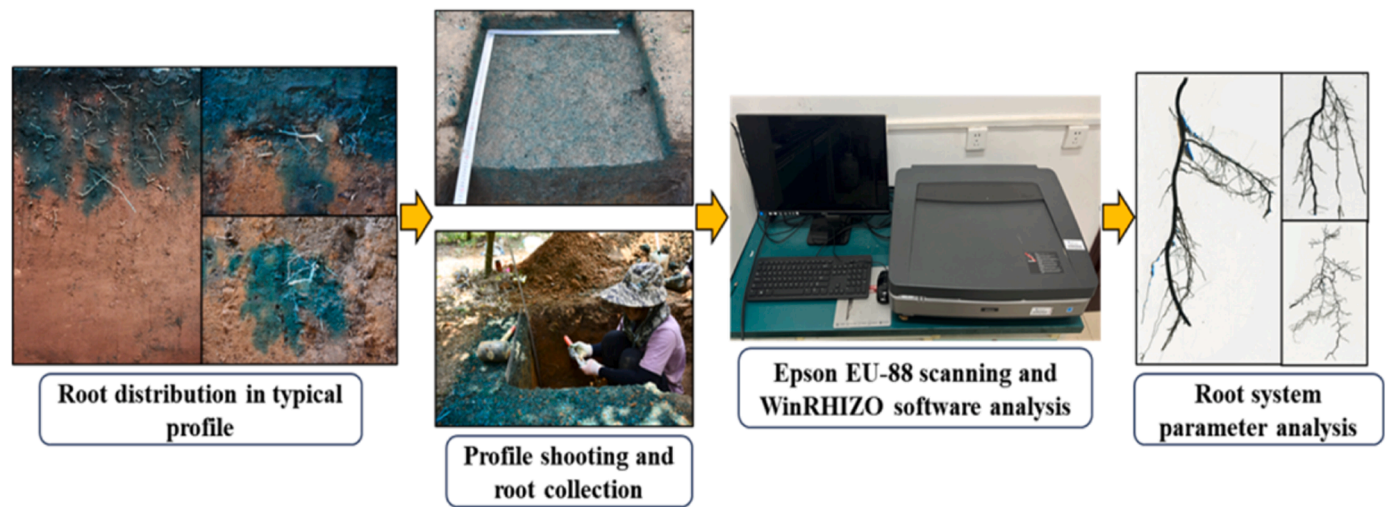


Fig. 4. Process of root acquisition and analysis in typical profiles.

Table 3
Root distribution parameters of soil vegetation with two lithologies.

Root parameters	Depth of soil layer/(cm)	Dolomite				Sandy shale			
		CL	OR	SG	SF	CL	OR	SG	SF
Root length density /(cm/dm^3)	0–10	132.65 $\pm 42.88\text{a}$	500.60 $\pm 36.78\text{a}$	952.04 $\pm 129.21\text{a}$	1255.63 $\pm 255.85\text{a}$	157.34 $\pm 33.20\text{a}$	2414.97 $\pm 471.85\text{a}$	1584.13 $\pm 57.76\text{a}$	3271.67 $\pm 193.08\text{a}$
	10–20	46.40 $\pm 7.81\text{b}$	307.08 $\pm 93.39\text{a}$	180.09 $\pm 15.98\text{b}$	211.20 $\pm 14.31\text{b}$	30.74 $\pm 3.66\text{b}$	716.36 $\pm 11.43\text{b}$	278.67 $\pm 51.98\text{b}$	825.68 $\pm 110.14\text{b}$
	20–30	49.82 $\pm 9.09\text{b}$	43.57 $\pm 23.87\text{b}$	106.07 $\pm 10.01\text{b}$	161.64 $\pm 19.12\text{b}$	31.13 $\pm 13.97\text{b}$	108.52 $\pm 23.16\text{c}$	276.88 $\pm 40.34\text{b}$	613.40 $\pm 54.01\text{bc}$
	30–40	6.13 $\pm 3.05\text{c}$	7.83 $\pm 1.86\text{b}$	36.43 $\pm 17.15\text{c}$	151.87 $\pm 1.24\text{b}$	30.52 $\pm 1.63\text{b}$	26.39 $\pm 12.75\text{d}$	246.64 $\pm 20.86\text{b}$	248.61 $\pm 13.33\text{cd}$
	40–50	————	12.46 $\pm 1.36\text{b}$	6.96 $\pm 1.14\text{c}$	62.97 $\pm 8.58\text{c}$	21.75 $\pm 1.63\text{b}$	60.16 $\pm 3.70\text{c}$	73.20 $\pm 4.87\text{c}$	164.75 $\pm 26.53\text{d}$
Root surface area density /(cm^2/dm^3)	0–10	29.44 $\pm 6.57\text{a}$	247.22 $\pm 12.38\text{a}$	218.82 $\pm 31.99\text{a}$	130.50 $\pm 10.09\text{a}$	110.94 $\pm 36.23\text{a}$	423.04 $\pm 87.99\text{a}$	430.20 $\pm 52.05\text{a}$	607.10 $\pm 156.25\text{a}$
	10–20	8.44 $\pm 6.34\text{b}$	72.28 $\pm 4.95\text{b}$	41.36 $\pm 1.66\text{b}$	111.30 $\pm 26.16\text{ab}$	31.22 $\pm 11.56\text{b}$	125.80 $\pm 25.53\text{b}$	88.62 $\pm 12.20\text{b}$	163.39 $\pm 20.44\text{b}$
	20–30	9.37 $\pm 0.90\text{b}$	8.79 $\pm 5.06\text{c}$	24.26 $\pm 0.76\text{b}$	66.91 $\pm 4.11\text{bc}$	25.21 $\pm 11.52\text{b}$	26.39 $\pm 9.35\text{c}$	58.42 $\pm 7.26\text{b}$	131.73 $\pm 9.39\text{b}$
	30–40	1.55 $\pm 0.37\text{c}$	2.52 $\pm 0.43\text{c}$	6.32 $\pm 2.21\text{c}$	63.32 $\pm 7.30\text{bc}$	12.03 $\pm 4.39\text{b}$	5.53 $\pm 2.28\text{d}$	48.34 $\pm 4.44\text{b}$	54.27 $\pm 18.88\text{c}$
	40–50	————	2.97 $\pm 0.41\text{c}$	1.08 $\pm 0.19\text{c}$	40.53 $\pm 19.20\text{c}$	12.44 $\pm 4.86\text{b}$	11.32 $\pm 1.29\text{c}$	16.32 $\pm 1.74\text{c}$	46.46 $\pm 14.71\text{c}$
Root volume density /(cm^3/dm^3)	0–10	0.56 $\pm 0.05\text{a}$	4.94 $\pm 0.54\text{a}$	5.52 $\pm 1.10\text{a}$	4.28 $\pm 1.37\text{ab}$	1.10 $\pm 0.06\text{a}$	6.25 $\pm 1.24\text{a}$	9.61 $\pm 2.19\text{a}$	12.28 $\pm 1.86\text{a}$
	10–20	0.13 $\pm 0.08\text{b}$	1.39 $\pm 0.15\text{b}$	0.84 $\pm 0.07\text{b}$	7.06 $\pm 0.49\text{a}$	0.69 $\pm 0.16\text{ab}$	1.81 $\pm 0.60\text{b}$	2.69 $\pm 0.32\text{b}$	5.29 $\pm 2.31\text{b}$
	20–30	0.14 $\pm 0.00\text{b}$	0.14 $\pm 0.09\text{c}$	0.49 $\pm 0.05\text{b}$	2.53 $\pm 1.26\text{b}$	0.63 $\pm 0.23\text{ab}$	0.55 $\pm 0.21\text{c}$	2.41 $\pm 0.09\text{b}$	2.03 $\pm 0.91\text{c}$
	30–40	0.02 $\pm 0.01\text{c}$	0.03 $\pm 0.00\text{c}$	0.09 $\pm 0.02\text{c}$	2.81 $\pm 0.72\text{b}$	0.36 $\pm 0.11\text{b}$	0.25 $\pm 0.08\text{c}$	0.77 $\pm 0.09\text{c}$	1.49 $\pm 0.18\text{c}$
	40–50	————	0.06 $\pm 0.01\text{c}$	0.01 $\pm 0.00\text{c}$	3.90 $\pm 0.90\text{ab}$	0.31 $\pm 0.01\text{b}$	0.18 $\pm 0.05\text{c}$	0.32 $\pm 0.05\text{c}$	1.27 $\pm 0.67\text{c}$

Table 4
Criteria for distinguishing water flow types based on soil dyeing path width.

Flow type	Proportion of dyeing path width and area / (%)	
	SPW < 20 mm	SPW > 200 mm
Homogeneous matrix flow	< 20	> 60
Heterogeneous finger flow	< 20	30–60
High interaction large pore flow	< 20	< 30
Mixed interacting macropore flow	20–50	< 20
Low interaction large pore flow	> 50	< 20

uniform (Table 2). The OR and SG plots showed dyed path differentiation in the 7–11 cm soil layer, with a low degree of preferential flow differentiation; these profiles were characterized mainly by uniform infiltration and less obvious preferential flow characteristics. The DAR of the SF plot showed a "W"-shaped trend that spread through the entire profile, with a higher degree of preferential flow differentiation. The total dyed area of sandy shale decreased in the order of SF > OR > SG > CL. This indicated that SF had the best water infiltration capacity and that CL had the poorest water infiltration capacity for both soil types.

As shown in Fig. 8, the preferential flow in the two soils was unevenly distributed across the horizontal profile, with large differences in the DAR and standard deviation at the same level. The standard deviation of the DAR of the horizontal dolomite soil profile was significantly smaller than that of the sandy shale profile. This result indicated that the

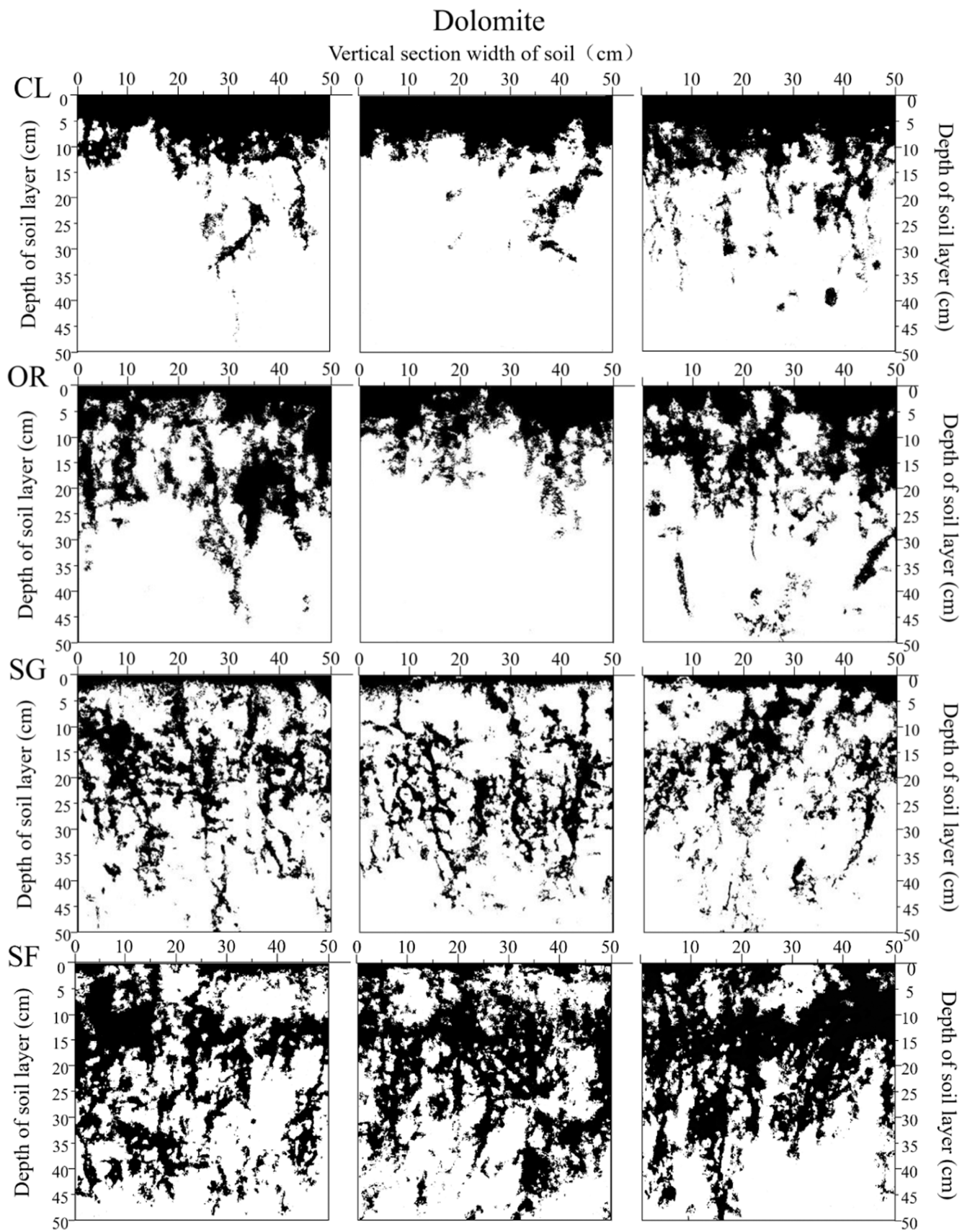


Fig. 5. Distribution characteristics of preferential flow patterns in the vertical profile of dolomite soil.

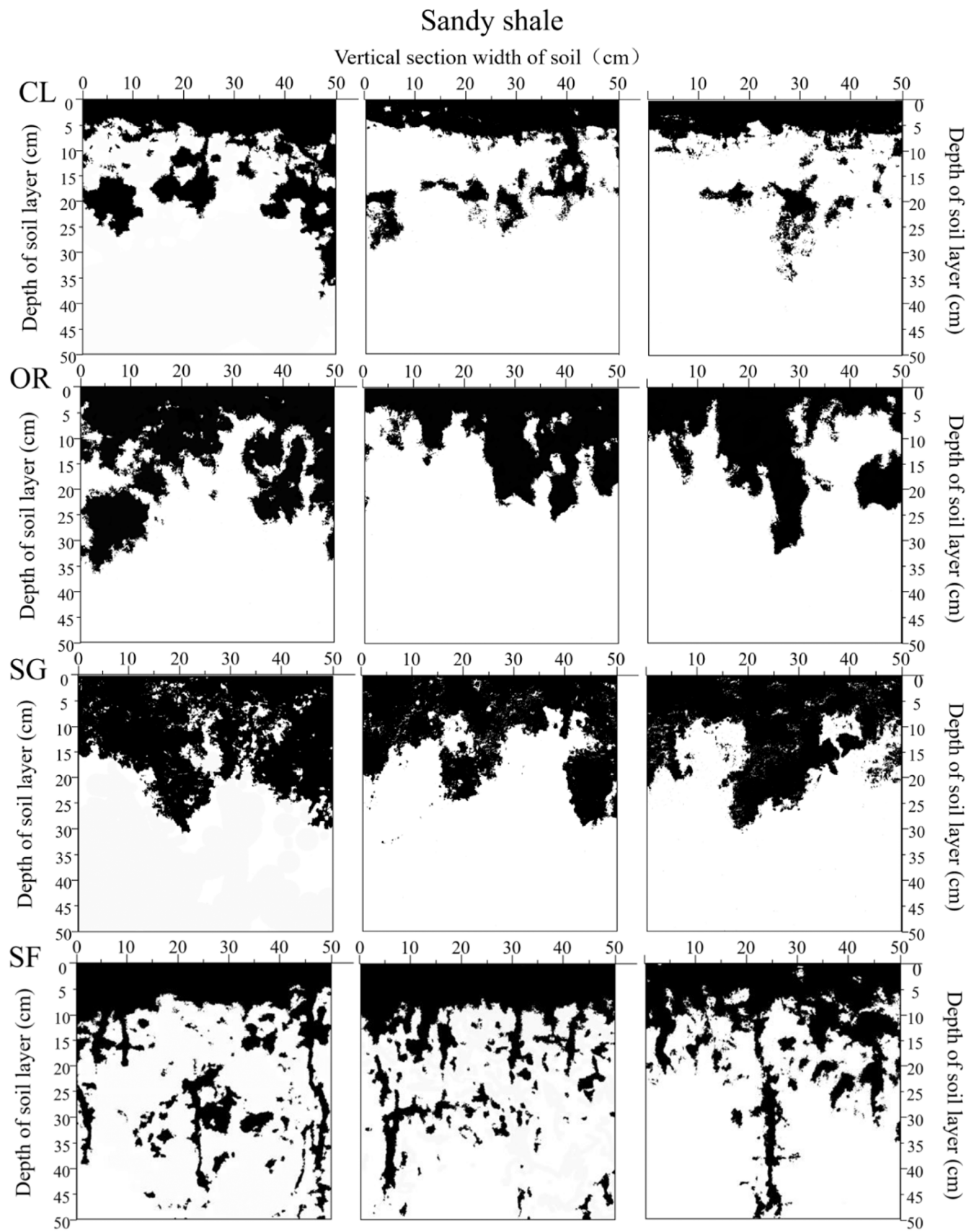


Fig. 6. Distribution characteristics of preferential flow patterns in the vertical profile of sandy shale soil.

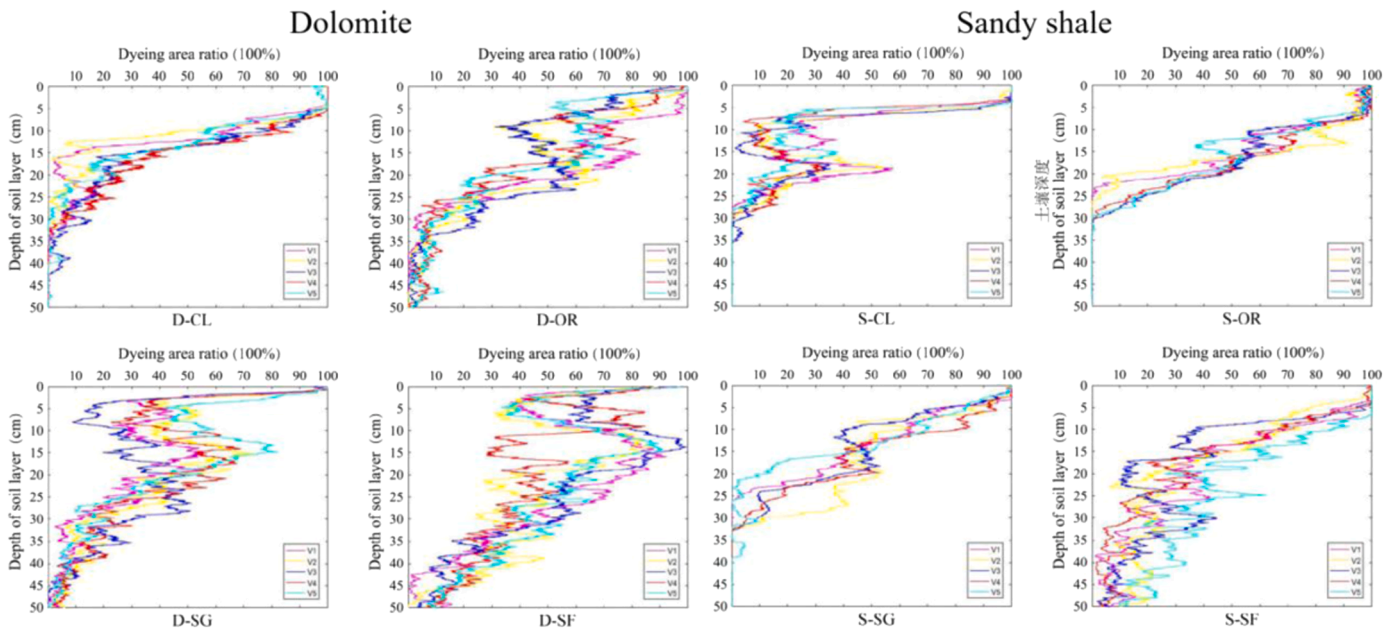


Fig. 7. Characteristics of dyeing area ratio in vertical profiles of two types of lithological developed soils.

Table 5
Ratio of dyeing area in vertical profile of dolomite soil.

Sample land type	Depth of soil layer/(cm)	Dolomite			Sandy shale		
		Average dyeing area ratio/(%)	Average total dyeing area /(cm ²)	Matrix flow depth/(cm)	Average dyeing area ratio/(%)	Average total dyeing area/(cm ²)	Matrix flow depth/(cm)
CL	0–10	92.28±2.83	683.34±88.70d	7.94±0.90Aa	64.33±4.80	486.99±43.96d	4.95±0.50Bc
	10–20	32.45±11.37			21.90±5.70		
	20–30	10.81±4.07			10.95±2.21		
	30–40	1.92±1.62			0.23±0.44		
	40–50	0.24±0.30			0.00±0.00		
OR	0–10	74.60±8.61	848.89±47.78b	3.04±1.32Bb	92.91±1.13	811.44±34.57ab	8.68±1.11Aa
	10–20	57.73±7.12			56.35±3.68		
	20–30	28.10±7.41			13.01±7.17		
	30–40	6.69±2.13			0.04±0.03		
	40–50	2.67±1.72			0.01±0.01		
SG	0–10	49.42±9.65	754.46±86.12c	1.70±0.34Bc	83.29±4.51	735.50±77.41c	6.98±0.64Ab
	10–20	50.36±11.26			45.29±3.29		
	20–30	33.02±6.67			17.17±11.97		
	30–40	13.30±4.45			1.38±0.83		
	40–50	4.80±1.01			0.00±0.00		
SF	0–10	57.89±9.14	1121.30±92.00a	1.27±0.32Bc	88.94±5.08	947.43±154.78a	8.79±0.87Aa
	10–20	70.71±15.89			45.01±10.58		
	20–30	49.56±9.55			26.32±8.03		
	30–40	32.88±4.57			17.77±7.97		
	40–50	13.20±4.01			11.47±5.91		

presence of more preferential pathways in the dolomite soils contributed to the occurrence of preferential flow. The standard deviation of the dolomite OR, SG, and SF DARs was small, while that of CL was large. This suggested that there was little variability in the distribution of the dyed paths, which also echoed the results of the vertical profile measurements.

3.3. Distribution characteristics of the dyeing path and water flow type

The vertical profile dyed paths and water flow characteristics in karst soil are shown in Figs. 9 and 10. The dyed paths of both soils were dominated by widths $W<20$ mm and produced multiple peaks with soil depth that were concentrated in the 5–20 cm soil layer. Among them, the dolomite path widths at all levels were significantly greater than those of sandy shale, indicating that more abundant soil dyed paths developed in dolomite. The number of $20\leq W\leq 200$ mm and $W>200$ mm

paths was low in both rocky soils, the distribution characteristics of the $20\leq W\leq 200$ mm and <20 mm dyed paths were more consistent, and the number of $W>200$ mm paths was the lowest, with most of these paths distributed in the 0–20 cm layer. This shows that there were more interconnected dyed paths in the shallow layer of karst soil, which allowed water transport to be uniform. With increasing soil depth, the large pores and cracks formed by structural development of soil decreased, and the number of preferential paths also decreased.

Water flow in the two rocky soils showed a transition from homogeneous matrix flow to heterogeneous finger-like flow and then to highly interactive macroporous flow (Fig. 10). The dolomite and sandy shale had high percentages of highly interactive macroporous flow, ranging from 55.11 % to 68.48 % and 13.37–67.83 %, respectively, and the percentages in the dolomite soil were greater than those in the sandy shale soil. The OR, SG and SF homogeneous matrix flow areas of the dolomite were obviously smaller than those of the sandy shale. Changes

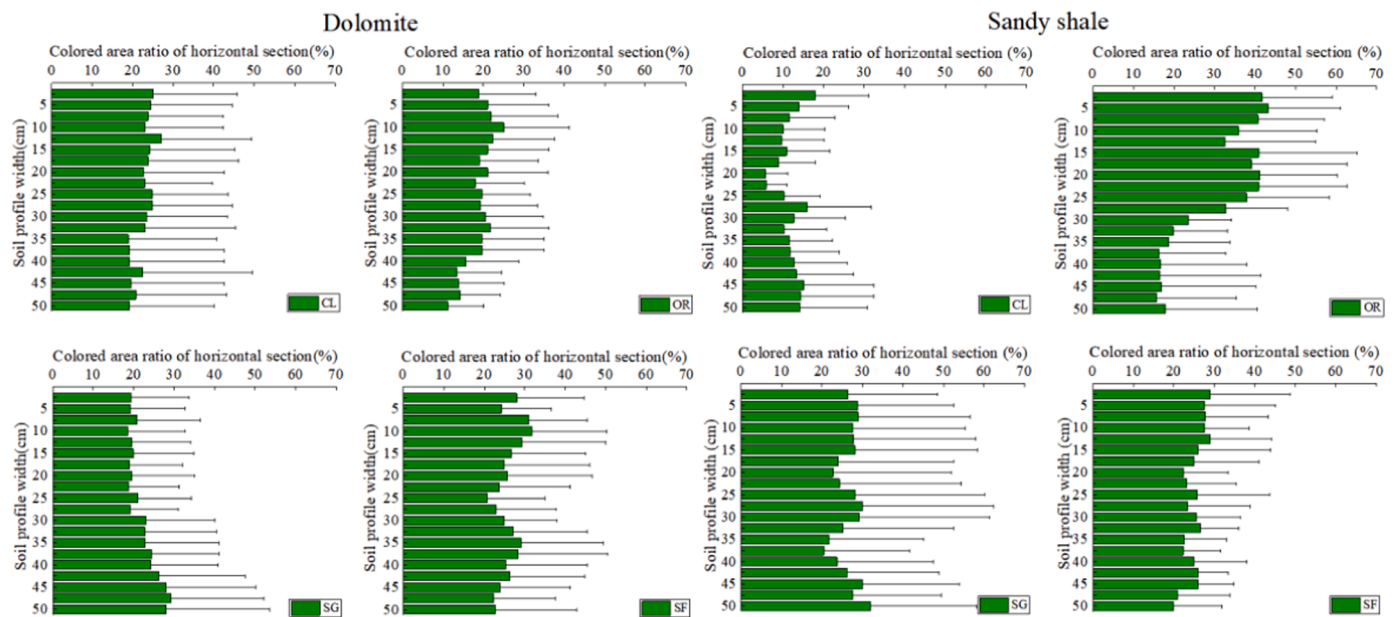


Fig. 8. Characteristics of dyeing area ratio in horizontal profiles of two types of lithology developed soils.

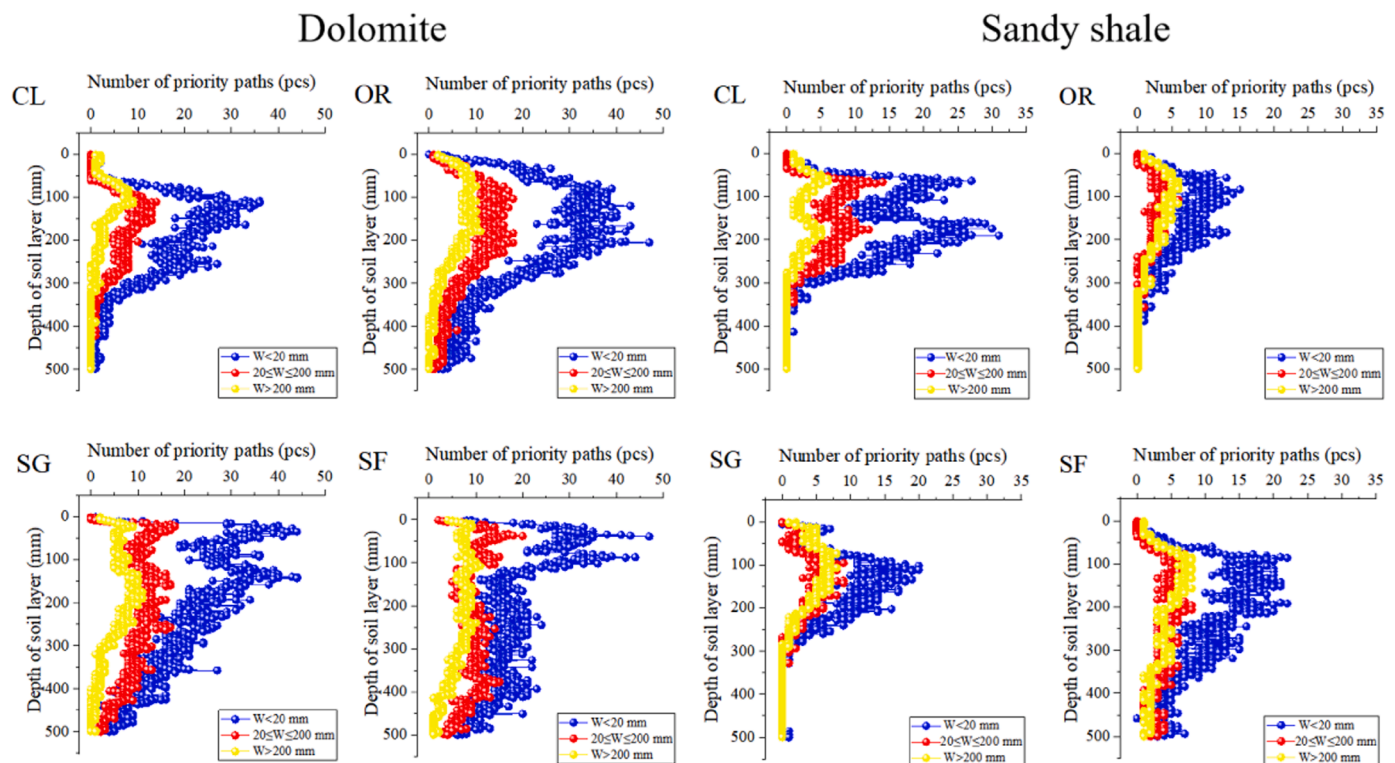


Fig. 9. Variation characteristics of dyeing paths in vertical soil profiles of two lithologies.

in land-use patterns also affected the extent of matrix flow. The proportion of homogeneous matrix flow in the CL and OR samples of dolomite was much greater than that in the SG and SF samples, reaching 25.79 % and 14.89 %, respectively, while the proportion of heterogeneous matrix flow was 5.73–30.60 %. The water infiltration depth of the SF of sandy shale was significantly greater than that of the SG, OR, and CL. The main type of water flow in the study area was high-interaction macroporous flow, and the land use type also had an impact on the water flow type.

3.4. Relationships between the characteristic parameters of root distribution and preferential flow

The pore channels formed by interactions between plant growth or dead roots and soil are important sources for the formation of preferential flow paths in soil. In this paper, the distribution characteristics of plant roots were fitted with the dyed area ratio of preferential flow (Fig. 11), and it was found that the two variables exhibited a power function relationship. The R^2 values for the root length density, root surface area density, and root bulk density fitted to the dyed area ratio

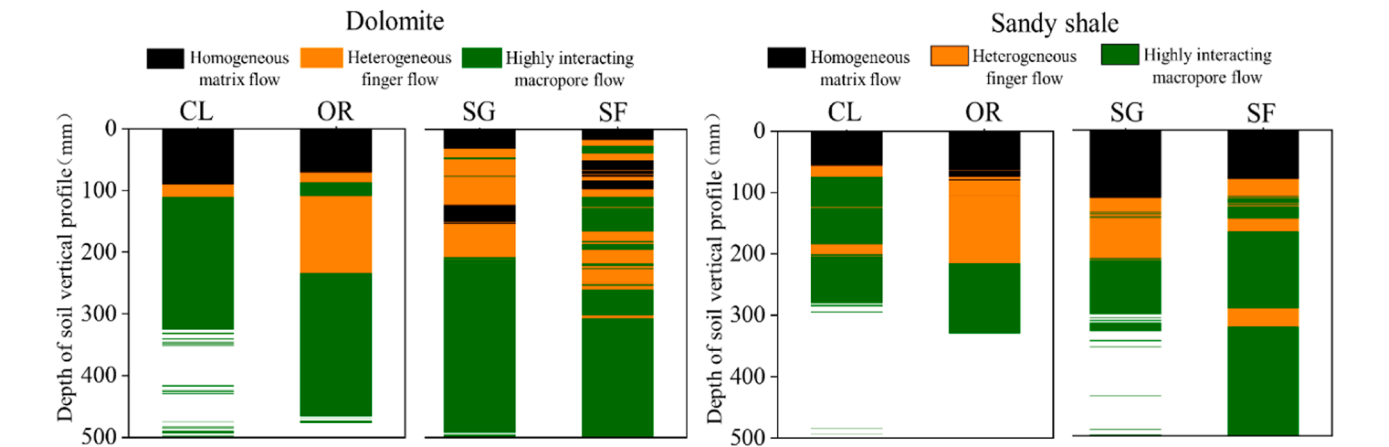


Fig. 10. Variation characteristics of two lithologic soil water flow types.

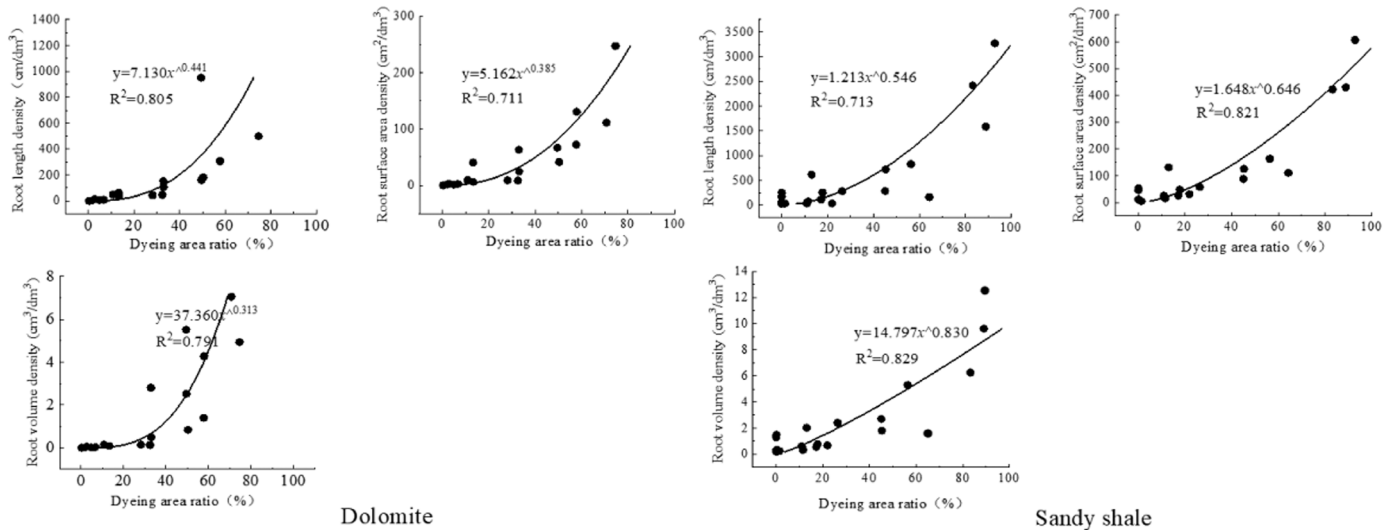


Fig. 11. Regression equation between root parameters and dyed area ratio of profile.

were 0.805 and 0.713, 0.711 and 0.821, and 0.791 and 0.829 for dolomite and sandy shale soil, respectively. These results indicated that plant root growth was effective at increasing the dyed area ratio and facilitating the occurrence of preferential flow. Correlation analysis between the number of dyed paths and the root parameters in the two rocky soils revealed that the number of dyed paths with $W > 200$ mm was significantly positively correlated with the root length density ($p < 0.05$) and was significantly positively correlated with the root surface area

density and root bulk density ($p < 0.01$) in the dolomite soil (Table 6). This result indicated that vegetative roots in dolomite soils influenced the characteristics of preferential flow mainly by affecting the number of $W > 200$ mm dyed paths.

The number of $W < 20$ mm dyed paths in the sandy shale soil was positively correlated with root length density, root surface area density and root bulk density, and the correlation coefficients were 0.504, 0.514 and 0.537 ($p < 0.05$), respectively. The number of dyed paths with

Table 6
 Correlation analysis between soil dyeing paths and root parameters of two lithologic soils.

Lithology	Correlation analysis	Root length density / (cm/dm ³)	Root surface area density / (cm ² /dm ³)	Root bulk density / (cm ³ /dm ³)	Preferential flow dyeing path / (pcs)		
					W < 20 mm	20 ≤ W ≤ 200 mm	W > 200 mm
Dolomite	Root length density	1	0.771**	0.639**	0.137	0.106	0.516*
	Root surface area density	0.771**	1	0.835**	0.210	0.196	0.649**
	Root bulk density	0.639**	0.835**	1	0.305	0.282	0.725**
	W < 20 mm	0.137	0.210	0.305	1	0.966**	0.765**
	20 ≤ W ≤ 200 mm	0.106	0.196	0.282	0.966**	1	0.723**
	W > 200 mm	0.516*	0.649**	0.725**	0.765**	0.723**	1
Sandy shale	Root length density	1	0.978**	0.929**	0.504*	0.388	0.583**
	Root surface area density	0.978**	1	0.965**	0.514*	0.379	0.593**
	Root bulk density	0.929**	0.965**	1	0.537*	0.420	0.612**
	W < 20 mm	0.504*	0.514*	0.537*	1	0.921**	0.809**
	20 ≤ W ≤ 200 mm	0.388	0.379	0.420	0.921**	1	0.807**
	W > 200 mm	0.583**	0.593**	0.612**	0.809**	0.807**	1

$W > 200$ mm was significantly correlated with the root length density (0.583), root surface area density (0.593) and root bulk density (0.612) ($p < 0.01$). Taken together, these findings indicate that vegetation roots affected preferential flow in soil mainly by influencing the number of $W < 20$ mm and $W > 200$ mm dyeing paths, and that the $W > 200$ mm dyeing path was the most affected.

3.5. Relationships between basic soil properties and preferential flow

A correlation analysis between preferential flow characteristics and soil properties is shown in Fig. 12. The water content, total porosity and organic matter content of the dolomite soil were positively correlated with the dyed area ratio and the number of dyed paths of each width, while the bulk density was negatively correlated ($p < 0.01$). These results indicate that the higher the water content of dolomite soil was, the greater the total porosity and the higher the organic matter content were, the larger the dyed area ratio was, the greater the number of dyed paths are and the greater the degree of preferential flow development was. In addition, there was a significant positive correlation between the sandy matter content and the dyed area ratio ($p < 0.05$), indicating that the soil particle size had an effect on preferential flow.

The water content of sandy shale soil was positively correlated with the ratio of the dyed area and the number of dyed paths with $W > 200$ mm but was not significantly correlated with the number of dyed paths with $W < 20$ mm and $20 \leq W \leq 200$ mm ($p < 0.01$). These results indicated that the soil water content could affect the number of dyeing paths and the development of the preferred flow. The soil bulk density was negatively correlated with the ratio of the dyed area and the number of dyed paths with $W > 200$ mm ($p < 0.05$), while the total porosity and organic matter content were positively correlated with the ratio of the dyed area and the number of dyed paths ($p < 0.01$). This shows that the lower the soil bulk density was, the greater the total porosity was, the greater the organic matter content was, the greater the dyed area ratio was, and the greater the number of dyed paths was; additionally, dyed paths with $W > 200$ mm had the most significant influence.

3.6. Comprehensive analysis of the formation factors of preferential flow

Soil preferential flow characteristics are often affected by a combination of several factors (Flury et al., 1994). In this study, on the basis of correlation analysis, we used characteristic parameters of preferential flow as the response variables and soil properties and characteristic parameters of root distribution as the end variables in the term

redundancy analysis to reveal the dominant factors affecting the preferential flow characteristics of the two types of rocky soils (Fig. 13). The results showed that the first-and second-ranking axes of the RDA cumulatively explained 76.16 % of the environmental factors influencing the preferential flow parameters of dolomite (Table 7). The first-ranking axis included the characteristic parameters of the preferential flow of the dyed area ratio and the number of dyed paths with different widths, which are referred to as the occurrence features of preferential flow and include the most effective factors influencing preferential flow. Among them, the sand content, total porosity, soil water content, organic matter, and plant parameters were positively correlated with the first ordination axis. The contribution of each factor was in the order of organic matter (69.10 %) > root length density (12.20 %) > soil water content (7.90 %) > root bulk density (3.50 %), which was significant ($p < 0.05$). This indicated that the organic matter content was the primary factor influencing the soil preferential flow characteristics of soil, followed by the root distribution. Therefore, the higher the organic matter content of dolomite soil was, the more developed and larger the root system was, the faster the uneven infiltration of soil water was, and the greater the development of preferential flow was.

According to the redundancy analysis of sandy shale, the first-order explanation rate was 87.08 %, which included most of the factors influencing preferential flow characteristics. The soil water content, total porosity, organic matter content and root distribution parameters were positively correlated with the first-ranking axis, and the density of the root surface area had the greatest contribution (77.20 %), followed by organic matter (11.90 %), sand (5.30 %), bulk weight (2.70 %), and soil water content (2.60 %). All of these factors were significant ($p < 0.05$). These results show that the density of the root surface area has had the greatest influence on the development of preferential flow in sandy shale soil, followed by the content of organic matter. In summary, there were many effective factors affecting preferential flow in the two rocky soils in the karst area, the dominant factors were different, and the soil water content had a large impact on preferential flow characteristics in the study area.

4. Discussion

4.1. Distribution of preferential flow in the two types of rocky soils

The distribution characteristics of preferential flow in the vertical soil profile can directly reflect dynamic changes in soil water infiltration. The DAR is an important quantitative index for measuring the

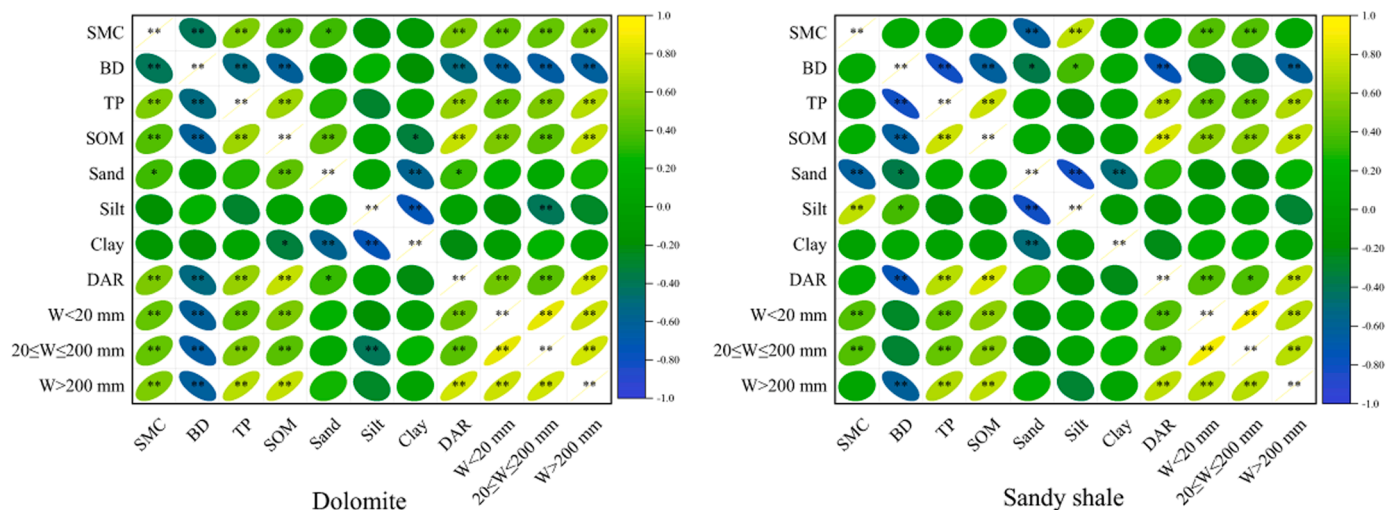


Fig. 12. Correlation analysis between two lithologic dyeing area ratios and basic soil properties. Note: SMC stands for soil water content; BD stands for soil bulk density; SOM stands for soil organic matter; DAR stands for dye area ratio. The same below.

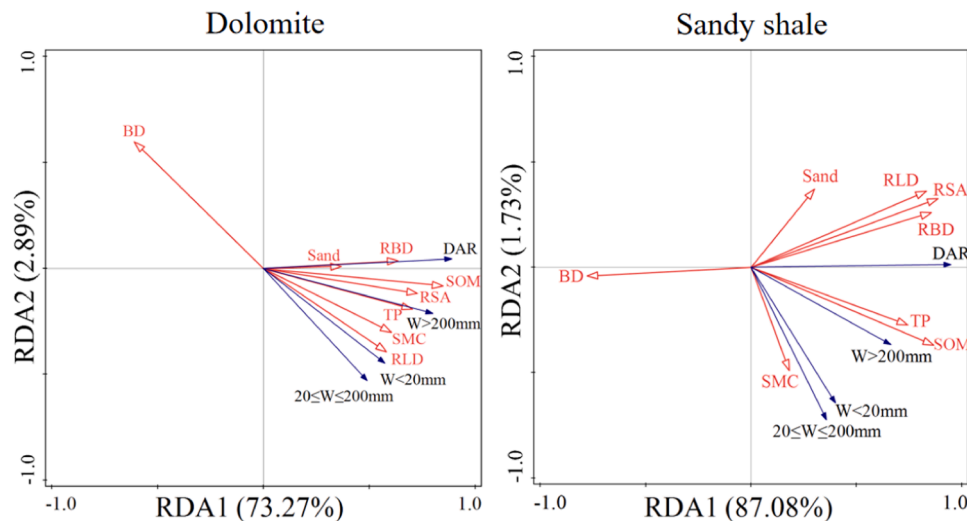


Fig. 13. Redundancy analysis of the characteristic parameters of soil preferential flow and environmental factors for two lithologies. Note: RLD represents root length density; RSA stands for root surface area density; RBD stands for root volume density.

Table 7
Redundancy analysis of influencing factors of soil priority flow.

Lithology	Influence factor	Explained variable/ (%)	Contribution rate/ (%)	F	P
Dolomite	SOM	52.70	69.10	42.40	0.002
	RLD	9.30	12.20	11.40	0.002
	SMC	6.00	7.90	5.40	0.016
	RBD	2.70	3.50	3.50	0.044
Sandy shale	RSA	68.60	77.20	83.10	0.002
	SOM	10.60	11.90	18.90	0.002
	Sand	4.70	5.30	14.10	0.002
	BD	2.40	2.70	4.70	0.040
	SMC	2.30	2.60	5.00	0.018

preferential transport process of water and solutes in the soil and reflects the total amount of transport (Tobella et al., 2014).

In this study, images of dyeing of two rocky soils from karst areas were analyzed, and the results showed that preferential flow was evident in both dolomite and sandy shale soils. The preferential flow paths in the dolomite soil mostly consisted of dendrites and some small patches, and the dyed paths were distributed in a network with high connectivity. Dyed path differentiation was observed in the shallow layer of the soil; this process was most significant in the SF and SG soils, followed by that in the OR soil, and the water in the CL soil mainly infiltrated in a homogeneous manner. The preferential flow pattern of sandy shale soil was mostly dominated by large-area clusters and blocks, and those in the OR and SG were characterized by finger-like and funnel-like features, with a large range of uniform infiltration between various land use types. In addition, the proportions of high-interaction macroporous flow and the numbers of dyed paths of different widths in the dolomite area were greater than those in the sandy shale area, but the depth of the soil matrix flow was much smaller in the dolomite area than in the sandy shale area. This result indicated that the degree of preferential flow development in the dolomite soil was greater than that in the sandy shale soil. This difference may be related to the unique binary structure of carbonate rocks. Studies have shown that carbonate areas are subject to aboveground erosion and subsurface dissolution of material, forming a special karst structure that contributes to the formation of many pores in the soil. These pores serve as natural drainage channels for mid-soil flow and promote the preferential flow of water into the bedrock, which further affects the development of subsurface karst conduits (Pauline et al., 2017) and can effectively promote the rapid generation

of preferential flow in dolomite soils. The DAR of dolomite showed an "S"-shaped pattern of change with increasing soil depth, while the DAR of sandy shale showed a linear decrease with increasing soil depth. This could be related to the presence of large pores and nonsoil components in the soils, such as transverse root systems, which can change the direction of water infiltration as well as soil porosity (Jiang et al., 2018, 2017) and inhibit vertical infiltration by transporting water horizontally (Chelli et al., 2018; Hardie et al., 2011). The preferential flow characteristics of sandy shale soils may be related to the soil texture. Related studies have shown that when the soil texture is coarse and the sand content is high, the soil tends to experience finger-like flow and funnel flow (Kung and Donohue, 1991; Yao and Hendrickx, 1996), which is consistent with research results showing that sandy shale soil has a higher sand content and a coarser texture than the dolomite soil (Fig. 6). Additionally, the sandy shale soil is classified as sandy loam and loamy soil, which are suitable for plant root growth and biological activities; as a result, the pore space in the soil and the capacity for material exchange between the matrix are high, and soil water transport exhibits obvious lateral infiltration and is dominated by homogeneous infiltration. Therefore, soil water infiltration in sandy shale areas exhibited more significant homogeneous transport and greater widths of dyed paths.

Differences in land use practices also affected the preferential flow characteristics of soils and altered their dyeing patterns. The degree of preferential flow differentiation and dyed path connectivity between the two rocky soils were greatest in SF and SG, followed by OR, and were the lowest in CL. This difference was mainly caused by the different frequencies of human disturbance caused by the difference in land use patterns and the difference in plant root distribution. Land use patterns can change the pore structural characteristics of soil (Tang et al., 2016). Perennial cultivation can accelerate the rate of mineralization of organic matter in soil and cause disruptions in pore connectivity, contributing to changes in water infiltration as well as in the mode of infiltration. In contrast, forested land with relatively little anthropogenic disturbance has abundant connected macropores (Wei et al., 2023). The pore fissures created by root growth and development can also promote preferential water transport to deeper soil. Therefore, the degree of preferential flow development in CL with a high frequency of anthropogenic disturbance was relatively low, and water transport was dominated by uniform infiltration.

4.2. Influence of plant roots and soil properties on preferential flow

Pore channels formed by interactions between the growing or dying

root system of plants and the soil are prerequisites for rapid water and solute transport, promoting preferential flow (Jørgensen et al., 2002). Related studies have shown that plant root systems are effective at improving the soil structure and reducing the bulk weight of soil (Bodner et al., 2008; Glab, 2013). The structure of the soil macroporous network in the soil formed by abundant plant root systems in the soil surface layer was the main basis for the formation of preferential pathways (Bodner et al., 2008; Tracy et al., 2011). The DAR is an important parameter for characterizing the development of preferential flow (Perillo et al., 1999). In this study, there was a power function relationship between the root distribution parameters and the DAR, with R^2 values ranging from 0.711 to 0.829. According to the correlation analysis, the root distribution parameters in the two areas with different lithologies were significantly positively correlated with the number of $W > 200$ mm preferential paths, while those in the sandy shale soil were significantly positively correlated with the number of $W < 20$ mm and $W > 200$ mm dyed paths ($p < 0.01$). The root distribution in karst areas affected the number and width of dyed pathways, which can contribute to the onset of preferential flow, in agreement with the findings of Murielle et al. (Murielle et al., 2011) and Bengough (Bengougha, 2012). Scholars have noted that water and solutes mainly exist in the root distribution area during infiltration (Bengougha, 2012), which indicates that the distribution characteristics of plant roots can affect the distribution of preferential paths and change morphology of the occurrence of preferential flows. In this study, the root distribution parameters in the area with sandy shale were greater than those in the dolomite area, which was due to the adequate moisture and temperature conditions in the study area, as well as the uneven spatial and temporal distributions of precipitation, frequent floods, extremely fast rates of water infiltration and poor water-holding capacity of carbonate rock-derived soils under the effect of a dichotomous structure (Pauline et al., 2017). Sandy shale soils are deeper and have better soil water-holding capacity, making them more suitable for plant root growth. Plant root systems concentrated in the surface layer of sandy shale soils form a complex network of large pores (Bodner et al., 2008), which contributes to the predominance of water infiltration over a large, uniform area, with significant lateral infiltration (Fig. 6) and a relative lag in preferential flow differentiation. Therefore, the difference in root distribution characteristics may be an important factor leading to the observed differences in preferential flow patterns between the two rocky soils.

In addition, differences in soil physical and chemical properties lead to differences in water infiltration characteristics. According to the correlation analysis, the soil water content, total porosity and organic matter content of dolomite were positively correlated with parameters of preferential flow development, and the bulk density was negatively correlated with parameters of preferential flow development ($p < 0.01$). These results indicated that the higher the soil water content was, the greater the total porosity and organic matter content were, and the lower the bulk density was, the greater the degree of main flow was. In sandy and shale soils, the lower the bulk density was, the greater the total pore number and organic matter content, the greater the degree of preferential flow development and the greater the number of dyed paths. The soil moisture content was found to contribute to preferential water flow, which is consistent with the findings of Jarvis and others (Jarvis et al., 2007) and is also recognized by many experts (Greve et al., 2010; Hardie et al., 2011). Researchers believe that this occurs because wet soil can inhibit the lateral infiltration of water and promote its rapid transport to deep soil layers (Flury et al., 1994). Moreover, a higher water content can effectively reduce the interaction between water and the soil matrix, increase infiltration depth, and increase the proportion of macroporous flow (Greve et al., 2010; Jarvis et al., 2007). However, the soil water contents of sandy shale were significantly lower than those of dolomite and were positively but not significantly correlated with the DAR, which may be related to the well-developed plant root systems in the soil.

4.3. Main formation factors of preferential flow in the two rocky soils

The nature of a soil, and the distribution of plant roots subtly influence the preferential flow characteristics of the soil (Allaire et al., 2009). To investigate the mechanism of preferential flow in soil in two typical rocky soils in the study area and to quantify the degree of influence of each environmental factor on preferential flow in soil, we used redundancy analysis to reveal the dominant factors driving soil preferential flow in soil in the two rocky soils. The results showed that the root surface area density was the dominant factor influencing preferential flow in sandy shale soils, and the contribution reached 77.20 %. The reason for this influence is that pores formed by root growth and decay are major sources of preferential pathways and can promote the creation of soil macropores (Vannoppen et al., 2015). Hydrological observations have revealed that pore space due to root penetration is a natural conduit for the direct transport of trapped precipitation to deeper soil layers, can create temporary saturated soil aquifers and can promote preferential flow (Devitt and Smith, 2002; Newman et al., 1998). Therefore, preferential flow in sandy shale soil, which has more developed plant root systems than dolomite soil, was influenced by plant root systems to a greater extent. Soil properties were also important factors influencing preferential flow characteristics, and the cumulative contribution of the organic matter content, bulk weight, soil moisture content, and sand content of sandy shale soils to preferential flow reached 22.50 %.

The organic matter content was the dominant factor driving preferential flow in the dolomite soil, and the contribution reached 69.10%. Studies have shown that the high surface cover of rock outcrops in carbonate ecosystems can lead to the accumulation of litter in soil patches (Göransson et al., 2014), increasing nutrient concentrations. Furthermore, relevant studies have shown that the number of soil microorganisms in karst areas is greater than that in nonkarst areas (Fan et al., 2018), and these microorganisms can improve the nutrient status in soil and promote nutrient cycling (Pan et al., 2018). Therefore, the content of organic matter in dolomite soil is significantly greater than that in sandy shale. Moreover, an increase in the soil organic matter content promotes the formation of stable soil structures and significantly increases soil porosity and permeability (Zhao et al., 2020). Soil organic matter, as a major source of nutrients, promotes plant growth, and plant roots sequester carbon; both of these processes are mutually beneficial and contribute to the further development of pore space in soil (Dal et al., 2012). Biological activity in soil also leads to an increase in the number of macropores. Thus, dolomite soils with high organic matter contents have dominant preferential flow factors that distinguish them from sandy shale soils. The influence of root distribution parameters on preferential flow is also relatively large, reaching 15.70%, reflecting the influence of plant root systems on preferential flow in dolomite soil. In summary, there are differences in the preferential flow characteristics of dolomite and sandy shale soils and in the main factors influencing the two in karst areas. The results presented herein provide a theoretical basis for the future management of rocky desertification as well as land-use planning in karst areas.

5. Conclusion

In this paper, a bright blue dyeing tracer combined with image processing techniques was used to observe and compare the distribution characteristics of preferential flow and the variability in factors influencing it in soil profiles developed from dolomite and sandy shale in the southwestern karst area of China. The results showed that preferential flow occurred in both dolomite and sandy shale soils in karst areas. The preferential flow morphology of the dolomite soil was mostly in the form of dendrites and small patches, with strong vertical water transport, a high degree of preferential flow development and strong spatial variability. In sandy shale soil, lateral water infiltration was obvious, and the

preferential flow pattern was dominated by clumps and blocks, with a low degree of preferential flow development and a large depth of matrix flow. Both the dolomite and sandy shale soils had high proportions of high-interaction macroporous flow, and the width of the dyed paths was dominated by $W < 20$ mm, with the number of dyed paths of different widths in the dolomite soils being greater than that in the sandy shale soils. Plant root distribution and soil properties affected the distribution characteristics of the ratio of the preferential flow area, and dyed paths with $W > 200$ mm were most affected. Redundancy analysis revealed that the development of preferential flow in soils of different lithologies was influenced by a variety of factors, with the main factor driving preferential flow in dolomite soils being the soil organic matter content and that in sandy shale soils being the root surface area density.

CRediT authorship contribution statement

Jianguo Wang: Formal analysis, Data curation. **Qingmei Meng:** Visualization, Validation, Software. **Yusong Deng:** Supervision, Funding acquisition. **Yuanfeng Yang:** Investigation. **Hui Wei:** Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of Competing Interest

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described is our original research that has not been published previously, nor under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the submission of the manuscript enclosed.

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